Non-Repeating Past Paradigm (PNP)

Backward travel selects a macro-compatible, but micro-distinct past; $\boldsymbol{Age_\Delta}$ remains monotonic, so classical self-erasure paradoxes do not obtain.

I. Core claim

A "**travel to the past**" **cannot land in the exact microstate that constituted one's own lived history.** Operationally, return to the identical past has measure zero. **Any backward jump leads to a compatible, but distinct past.** Classical self-nullification paradoxes therefore do not arise.

II. Assumptions

1 **Time as change:** Replace coordinate time with an orientation-free measure of accumulated change, Δ. The operational "age," Age_Δ is monotone non-decreasing along any trajectory, irrespective of expansion/implosion flips.

2 **Macro-micro multiplicity:** A macroscopic epoch t^* corresponds not to a single microstate but to a set $\mathcal{M}(t^*)$ of micro-configurations consistent with the same macroscopic constraints (macro-level variables).

3 **Measure-zero identity:** Selecting exactly the microstate $s \in \mathcal{M}(t^*)$ that coincided with one's original worldline requires perfect synchronization over all degrees of freedom; its probability is effectively zero.

III. Mechanism of "backward travel"

A backward transition samples some $s' \in \mathcal{M}(t^*)$ with $s' \neq s$ almost surely. Because Age_Δremains monotone, accumulated change is not undone; **the system is relocated to a different branch compatible with the target macro-constraints**. **Meeting "oneself" is possible only as encountering a counterpart on s', not the same Δ-trajectory.**

IV. Consequences

1 **No self-erasure paradox:** Killing a younger "you" in s'does not remove the traveler, who persists on their original Δ-trajectory.

2 **No causal loops that reset one's own history:** Monotonic Age_Δprecludes constructing a consistent loop that decreases accumulated change.

3 **Identity as trajectory:** Personal identity is a continuity in Δ, not a point event. Branching yields distinct, Δ-continuous agents rather than one agent duplicated.

V. Intuition

Each macroscopic "moment" is a thick equivalence class of microstates. Without complete control of the full micro-configuration, any attempt to re-enter a past macro-epoch almost surely lands in a **micro-distinct** member of that class. That is sufficient to dissolve standard time-travel paradoxes without special "self-consistency" postulates.

Summary. In a Δ-based framework where "time" is accumulated change and macroscopic epochs are many-to-one over microstates, backward travel selects a different compatible past rather than one's own. **Hence the past is non-repeating in practice, and classical paradoxes (kill-your-younger-self = vanish) do not obtain.**

VI. Additional example for the paradigm

Imagine you embark on a journey through time. According to this non-repeating past paradigm, when you return to what you thought was your original starting point, on a macro scale it might look the same. **The city, the house, the general setting all seem familiar, but on a micro level, details have shifted**. A person who was previously a close partner may now barely recognize the returning agent or even display hostility. The relationship you had is altered because the timeline you came back to is not an exact microstate replica of the one you left.

In other words, you can return to a scenario that's structurally similar, but the personal connections and minor elements have reconfigured. So that's the add-on example: **no need for the dramatic "kill your younger self" construct**. Instead, it's about returning to a slightly shifted reality where the fine details of relationships or personal histories have changed.

VII. Symbols used

1 t^*

A macroscopic epoch. Not a single frozen frame, but a label for "the world as described by a set of large-scale conditions" (distribution of matter, background temperature, etc.).

2 $\mathcal{M}(t^*)$

The set of all **microstates** that are compatible with that macroscopic epoch. In other words, many different fine-grained configurations that all look the same at the macro level.

3 $s \in \mathcal{M}(t^*)$

The particular microstate that the original worldline actually passed through at that epoch. This is the one that is practically impossible to hit again.

4 $s' \in \mathcal{M}(t^*)$

A different microstate from the same macro-class. This is the microstate into which the backward operator will almost certainly place the system when the return occurs, macro-consistent, micro-distinct.

So the consistent wording should be:
a) "Selecting exactly the microstate $s \in \mathcal{M}(t^*)$ that coincided with one's original worldline requires perfect synchronization over all degrees of freedom; its probability is effectively zero."
b) And, for the actual backward jump: "In practice the backward operator lands in $s' \in \mathcal{M}(t^*)$ with $s' \neq s$".

5 monotonicity of Age_Δ

a) Δ is the orientation-free measure of change.
b) Age_Δ is the accumulated change along the agent's trajectory.

"Monotonic" here simply means: Age_Δ(after the jump) $\geq$ Age_Δ(before the jump)
No operation of "time travel" is therefore allowed to decrease the already accumulated change. A transition can relocate the system to another compatible past, but it cannot unwind the existing Δ-history. This is what blocks the standard time-travel paradoxes.

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